Correspondence

Some Amendments to "Field-to-Wire Coupling in an Electrically Large Cavity: A Semianalytic Solution"

Huabin Zhang, Xiang Zhao, Liping Yan, Changjun Liu, and Kama Huang

***Abstract*—In this correspondence, we propose a supplementary spatial correlation function used to amend a simplification made in an earlier published paper in which a semianalytic solution of field-to-wire coupling in an electrically large cavity was first proposed. It can be shown that the horizontal and vertical components of the incident field are correlated in most cases. Without neglecting the correlation between the field components, the supplementary function can help to give more accurate results. It is demonstrated by two numerical simulation examples in which the semianalytic results are compared with the Monte Carlo results and/or the analytic solution.**

***Index Terms*—Amendments, corrections, field-to-wire, semianalytic solution, spatial correlation.**

I. Introduction

In the above paper [1], a semianalytic and statistical solution of field-to-wire coupling in an electrically large cavity was first proposed. Based on the Baum–Liu–Tesche (BLT) equation and the electromagnetic (EM) environment represented as a random plane-wave spectrum and characterized by its statistical parameters, the method evaluated the EM coupling to a wire running parallel to the wall of an oversized cavity. To this end, the statistical properties of EM fields near the walls were applied and several important correlation functions near the walls were presented. With the aid of these correlation functions, fifteen terms appearing on the right-hand sides of formulas (15), (17) and (19) in [1] were obtained to calculate the interesting parameters such as the mean of the current amplitude, the mean of the square of the current and the variance of the induced voltage at the end of the wire. It was assumed in [1] that the horizontal and vertical components of the incident field were independent. So, for example, several terms such as $\left\langle Q_y^1 Q_z^* \right\rangle$, $\left\langle Q_z Q_y^{1*} \right\rangle$, $\left\langle Q_z Q_y^{2*} \right\rangle$ and $\left\langle Q_y^2 Q_z^* \right\rangle$ were neglected in (15), but in fact these terms are not absolutely equal to zero, and are comparable to the other terms. It may further lead to some inaccurate results by neglecting the contributions of these terms.

In this correspondence, we propose a supplementary spatial correlation function to amend this problem. In Section II, the spatial correlation functions between the horizontal and vertical components of the incident field far from as well as near the wall are given, respectively. The functions can be used to calculate the neglected terms. Additionally, formulas (9) and (21) in [1] are inaccurate due to typing errors and are corrected in Section III. Section IV presents validations by two numerical simulation examples to support the corrected semianalytic solution that includes those neglected terms completely.

Manuscript received August 3, 2011; revised November 7, 2011; accepted December 10, 2011. Date of publication January 11, 2012; date of current version February 17, 2012. This work was supported by National Natural Science Foundation of China under Grant NSFC-60801035 and Grant NSAF-11176017.

H. Zhang is with Sichuan University, Chengdu 610065, China (e-mail: binbinstar-999@163.com).

X. Zhao, L. Yan, C. Liu, and K. Huang are with the School of Electronics and Information Engineering, Sichuan University, Chengdu 610065, China (e-mail: zhaoxiang@scu.edu.cn; liping_yan@scu.edu.cn; cjliu@ieee.org; kmhuang@scu.edu.cn).

Digital Object Identifier 10.1109/TEMC.2011.2180910

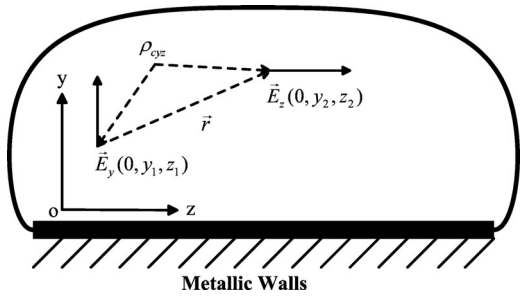


Fig. 1. Geometry for correlation function between the horizontal and vertical components of the incident field in the $y - o - z$ plane.

II. A Supplementary Spatial Correlation Function

Without loss of generality, at first, we define as (1) the spatial correlation function $\rho_{\mathrm{cyz}}(y_1, z_1, y_2, z_2)$ between two rectangular components $\vec{E}_y(0, y_1, z_1)$ located at $(0, y_1, z_1)$ and $\vec{E}_z(0, y_2, z_2)$ located at $(0, y_2, z_2)$ in a cavity, as shown in Fig. 1.

$$\rho_{\mathrm{cyz}}(y_1, z_1, y_2, z_2) = \frac{\langle E_y(0, y_1, z_1) E_z^*(0, y_2, z_2) \rangle}{\sqrt{\left\langle |E_y(0, y_1, z_1)|^2 \right\rangle \left\langle |E_z^*(0, y_2, z_2)|^2 \right\rangle}}. \quad (1)$$

For the case of observation points being far from the wall, we have

$$\rho_{\mathrm{cyz}}(y_1, z_1, y_2, z_2) = (y_2 - y_1)(z_2 - z_1) \big/ r^2 [\rho_l(r) - \rho_t(r)] \quad (2)$$

where $r = \sqrt{(y_1 - y_2)^2 + (z_1 - z_2)^2}$, $\rho_l(r)$, and $\rho_t(r)$ are given as follows [2]:

$$\rho_l(r) = 3 \big/ (k^2 r^2) \left[\sin(kr)/(kr) - \cos(kr)\right] \quad (3)$$

$$\rho_t(r) = 3/2 \left[\sin(kr)/(kr) + \cos(kr) \big/ (k^2 r^2) - \sin(kr) \big/ (k^3 r^3)\right] \quad (4)$$

in which k is the wavenumber in medium.

When the observation points are near the walls, proximity effects have to be considered. In this case, we use notation ρ_{cwyz} instead of ρ_{cyz}. Using the derivation method similar to [3], we obtain

$$\begin{aligned}\rho_{\mathrm{cwyz}}(y_1, z_1, y_2, z_2) = \{&(y_2 - y_1)(z_2 - z_1) \left[\rho_l(r_1) - \rho_t(r_1)\right] / r_1^2 \\ &+ (y_2 + y_1)(z_2 - z_1) \left[\rho_l(r_2) - \rho_t(r_2)\right] / r_2^2 \} \\ &/ [\sqrt{1 + \rho_l(2y_1)} \sqrt{1 - \rho_t(2y_2)}] \end{aligned} \quad (5)$$

where $r_1 = \sqrt{(y_1 - y_2)^2 + (z_1 - z_2)^2}$, and $r_2 = \sqrt{(y_1 + y_2)^2 + (z_1 - z_2)^2}$. As can be seen in (5), ρ_{cwyz} is obviously dependent on the two coordinates. For the case of $z_1 = z_2$, ρ_{cwyz} absolutely equals zero. But in many other cases, ρ_{cwyz} does not equal zero and decays in an oscillatory manner as kr increases. The limit of (5) for large ky_1 and ky_2 is just the same as (1). As an example, some values of ρ_{cwyz} with different coordinates and different wavenumbers are calculated, as shown in Fig. 2.

With the aid of (5), we can use the similar method, as shown in formula (16) in [1], to calculate the terms such as $\left\langle Q_y^1 Q_z^* \right\rangle$, $\left\langle Q_z Q_y^{1*} \right\rangle$, $\left\langle Q_z Q_y^{2*} \right\rangle$ and $\left\langle Q_y^2 Q_z^* \right\rangle$, and so on.

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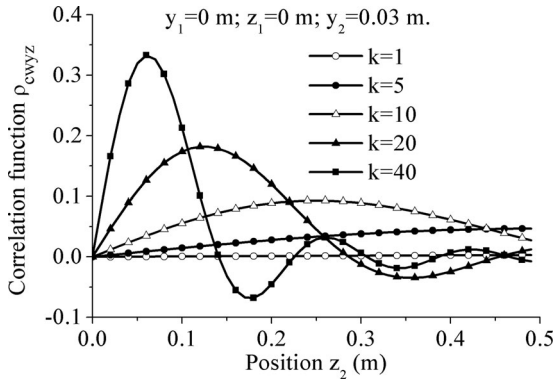


Fig. 2. Correlation function between the horizontal and vertical components of the incident field for the observation points near the walls, with increasing wavenumbers and different coordinates.

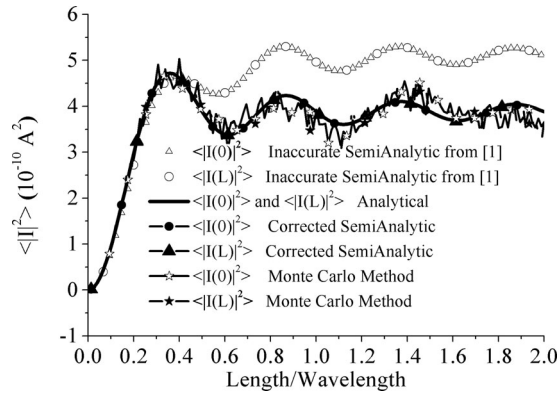


Fig. 3. Comparisons among the solutions.

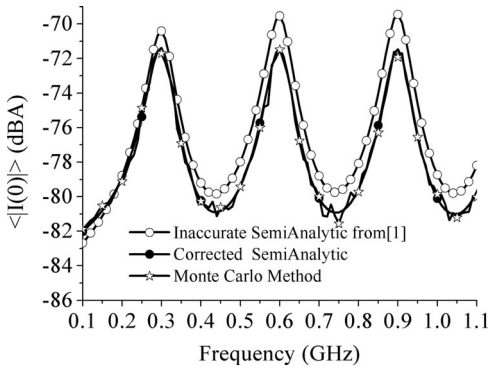


Fig. 4. Comparisons among the means of the current amplitude calculated by different methods.

III. Some Corrections

Formula (9) in [1] is inaccurate and is corrected as follows using the same derivation method [1], [3]:

$$\rho_{\mathrm{cw}}^{y}(y,\Delta y,\Delta z)=\frac{\rho_c^{y}(\Delta y,\Delta z)+\rho_c^{y}(2y+\Delta y,\Delta z)}{\sqrt{1+\rho_L(2y)}\sqrt{1+\rho_L(2y+2\Delta y)}}. \tag{6}$$

And, the last inaccurate formula (21) in [1] is corrected as

$$\langle|I_1|\rangle=\sqrt{\left\langle|I_1|^2\right\rangle}\sqrt{\pi/4} \tag{7}$$

which can be validated by the Monte Carlo method detailed in [4].

IV. Validations

In the first example, we are interested in the mean square of the current amplitude $\left\langle|I(0)|^2\right\rangle$ or $\left\langle|I(L)|^2\right\rangle$ at the end of a 40-cm-long–uniform-lossless cable, whose diameter is 1 mm, situated at 5-mm above a perfectly conducting ground plane and loaded at both ends by matched loads. In this section, $I(0)$ and $I(L)$ are derived from the EM coupling equation, BLT equation [5], based on the transmission line theory. The BLT equation is

$$\left\{\begin{array}{c} I(0) \\ I(L) \end{array}\right\}=\frac{1}{Z_c}\left[\begin{array}{cc} 1-\rho_1 & 0 \\ 0 & 1-\rho_2 \end{array}\right]\left[\begin{array}{cc} -\rho_1 & \mathrm{e}^{\gamma L} \\ \mathrm{e}^{\gamma L} & -\rho_2 \end{array}\right]^{-1}\left\{\begin{array}{c} S_1 \\ S_2 \end{array}\right\} \tag{8}$$

where

$$\left\{\begin{array}{c} S_1 \\ S_2 \end{array}\right\}=\frac{1}{2}\left\{\begin{array}{l} \displaystyle\int_0^L \mathrm{e}^{\gamma z}E_z(0,h,z)dz+\int_0^h E_y(0,y,0)dy \\ \quad -\mathrm{e}^{\gamma L}\displaystyle\int_0^h E_y(0,y,L)dy \\ -\mathrm{e}^{\gamma L}\displaystyle\int_0^L \mathrm{e}^{-\gamma z}E_z(0,h,z)dz-\mathrm{e}^{\gamma L}\int_0^h E_y(0,y,0)dy \\ \quad +\displaystyle\int_0^h E_y(0,y,L)dy \end{array}\right\}$$

and

$$\rho_1=(Z_1-Z_c)/(Z_1+Z_c)\quad \rho_2=(Z_2-Z_c)/(Z_2+Z_c).$$

On the one hand, $\left\langle|I(0)|^2\right\rangle$ and $\left\langle|I(L)|^2\right\rangle$ are analytically obtained by using the same method detailed in [6]

$$\left\langle|I(0)|^2\right\rangle=\left\langle|I(L)|^2\right\rangle=\left(E_0^2/4\right)(h/Z_c)^2\left[2-\sin(2kL)/(kL)\right] \tag{9}$$

where E_0^2 is a constant [7] and set to 1 $(\mathrm{V/m})^2$, h denotes the distance from the cable to the ground plane, L the length of the cable, k the free-space wavenumber, Z_c the characteristic impedance of the transmission line, and Z_1 and Z_2 are the impedances of the loads.

On the other hand, semianalytical solutions of $\left\langle|I(0)|^2\right\rangle$ and $\left\langle|I(\mathrm{L})|^2\right\rangle$ are obtained separately by using the semianalytical method in [1] without and with the contributions of the supplementary spatial correlation function shown in formula (5).

All aforementioned solutions are compared with the solutions obtained by the Monte Carlo method [4], as shown in Fig. 3. It can be seen from Fig. 3 that except the inaccurate semianalytical solutions of [1] all other solutions agree well with each other.

In the second example, we calculate the mean of the current amplitude $\langle|I(0)|\rangle$ at the end of a 50-cm-long cable, whose diameter is 1 mm, situated at 3-cm above a perfectly conducting ground plane and loaded at both ends by 50 Ω. For this same problem, as shown in [1], the curves in Fig. 4 illustrate that only the corrected solutions agree well with the ones by the Monte Carlo method.

In the Monte Carlo method in both examples, the convergence was reached by considering 250 successive channel realizations, each realization being simulated by 1000 random plane waves.

According to the aforementioned examples, the deviation between the inaccurate semianalytical solutions and the corrected ones implies that the neglected terms in [1] should not be neglected, or errors will be introduced.

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